Early Identification of At-Risk GCE A/L Candidates in Sri Lanka with an Explainable Boosting Machine

Applied Machine Learning: Assignment Report

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Index number: [add]

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Source code: https://github.com/niwanthaik/al-early-warning-ebm

Live app: https://al-early-warning.redwave-d9ae0d05.eastasia.azurecontainerapps.io

Demo video: [link]

Abstract. About 36% of Sri Lankan GCE Advanced Level (A/L) candidates fail to meet the basic university-entry requirement of passing all three subjects. Using the full national 2020 results (290,285 candidates, validated against official statistics), this work predicts, *before results are released*, which candidates are at risk, from their subject combination, Common General Test (CGT) score, General English grade, age and gender. A machine-learning algorithm not covered in the lectures, the **Explainable Boosting Machine (EBM)**, a glass-box generalised additive model with pairwise interactions trained by cyclic gradient boosting, is compared with the taught models (logistic regression, decision tree, k-NN, random forest, XGBoost) and a LightGBM black-box reference. The EBM reaches a test ROC-AUC of **0.767** (95% CI 0.763–0.771). That significantly beats logistic regression, the decision tree, k-NN and random forest, and is within 0.005 of the best black box, while being fully interpretable and the least over-fitted model. Its learned shape functions agree with seven independent pieces of domain knowledge, including the 31 January school-entry cohort effect and the CGT exemption for repeat candidates. They are cross-checked with PDP, permutation importance, SHAP and LIME. An early-warning threshold catches 70% of at-risk students. Removing gender equalises at-risk detection between girls and boys at a cost of 0.023 AUC, a trade-off discussed together with re-identification risk and other ethical limits. A Streamlit application exposes predictions and explanations to teachers.

1. Problem Definition and Dataset Collection (15 marks)

1.1 Problem and relevance

The GCE A/L examination decides access to Sri Lanka's free state universities. To be considered by the University Grants Commission (UGC), a candidate must first pass all three subjects (grade S or above). In 2020, 35.6% of candidates who sat all three papers did not. Schools and tuition teachers usually learn who struggled only when results are published, months after the examination, when it is too late to help. The question here is whether information that is available **before results** can identify at-risk candidates early enough for supportive intervention (extra classes, counselling, advice on re-sitting or alternative pathways). The task is **binary classification**: `eligible = 1` if all three subjects were passed, otherwise 0.

1.2 Data source and collection

The data is the 2020 GCE A/L results sheet published by the Department of Examinations and compiled into one table by Amarasinghe (2021) on Kaggle (licence CC BY-NC-SA 3.0 IGO). Names, National Identity Card (NIC) numbers and examination index numbers were removed by the publisher; birth date and gender were decoded from the NIC before removal, which explains the 'Invalid nic error' entries. Because this is a community compilation rather than an official release, it was **validated against the official statistics** (Table 1). The qualification rate matches to 0.04 percentage points, and the dataset covers 96.2% of candidates who sat and 96.3% of those who qualified. Coverage is the same for both groups, and also by gender (96.3% of qualified females, 95.8% of qualified males), so the missing ~4% shows no sign of selection bias.

Table 1. Validation of the dataset against official 2020 figures (Department of Examinations; UGC Statistics 2021, Table 2.1).

	Official (Dept. of Examinations / UGC)	This dataset	Coverage / match
Candidates who sat	301,771	290,285	96.2%
Qualified for university entrance	194,297	187,038	96.3%
Qualification rate	64.39%	64.43%	matches
Female share of qualified	64.6% (125,653 / 194,366)	64.8%	matches

1.3 Features, target and size

The raw file has 337,553 rows and 19 columns. After cleaning, **290,285 candidates** remain, described by 51 model features (Table 2): 4 numeric, 3 nominal, 2 missingness flags and 43 subject indicators. Candidates chose from 61 subjects, forming 2,662 distinct combinations. The 20 most common cover 72% of students, while 2,242 combinations have fewer than 30 students. The target is moderately imbalanced (64.4% eligible).

Table 2. Features and target.

Feature	Type	Definition / handling
cgt_score	numeric 0–100	Common General Test score; 'Absent' → missing + flag
ge_grade	ordinal 0–4	General English grade F/S/C/B/A; 'Absent' → missing + flag
age_at_exam	numeric	2020 − birth year (from NIC); implausible ages (outside 15–70) → missing
birth_month	numeric 1–12	captures the school-entry cohort (31 January cut-off)
gender	nominal	female / male; NIC errors → 'unknown'
syllabus	nominal	new / old (2020 had both)
stream	nominal (6)	re-derived from the subject combination (not the leaked official column)
cgt_absent, ge_absent	binary	explicit missingness indicators (repeat candidates are exempt)
SUBJ_* (42) + SUBJ_OTHER	binary / count	which of the 61 subjects the candidate took
eligible (target)	binary	1 = passed all three subjects (A/B/C/S), the basic UGC requirement

1.4 Preprocessing

Table 3 lists the cleaning steps. Two deserve emphasis. First, a **leakage audit** found that the official `stream`, `Zscore` and rank columns are filled in only for candidates who passed. Each one separates the classes perfectly (103,246 of 103,247 non-eligible candidates have a blank stream), so a model using them would look perfect and be useless. They were removed, and the stream was **re-derived from the subject choices** using the UGC stream rules. This reproduces the official stream for 99.9% of candidates in the six standard streams. Second, **missingness is informative**: candidates without a CGT score passed at 71.3% against 63.6% for those with one, because repeat candidates are exempt from the CGT and General English papers. Missing values are therefore kept as explicit indicators rather than silently imputed.

Table 3. Preprocessing steps.

Step	Rows / columns affected	Reason
Remove candidates absent/withheld for any subject paper	47,268	target undefined
Drop leaked columns: official stream, Z-score, district/island rank	4 columns	populated only for passing candidates (perfect separators)
Drop the subject results sub*_r	3 columns	they define the target
Parse 'Absent' CGT / General English → missing + indicator	11.4% / 7.7%	missingness is informative
NIC decoding errors (birth date, gender) → missing	0.24%	imputed (median / 'unknown')
Implausible ages (outside 15–70) → missing	32	data-entry errors
Re-derive stream from subjects (UGC rules)	99.9% agreement	avoids leakage, keeps the information
Encoding	–	one-hot (LR, k-NN, DT, RF, XGB, LGBM); native nominal handling (EBM)
Imputation / scaling	–	median + standardisation inside the pipeline for LR and k-NN; trees and EBM keep missing values as their own bin
Data minimisation	day of birth, index, ranks	never used and stripped from every shared file

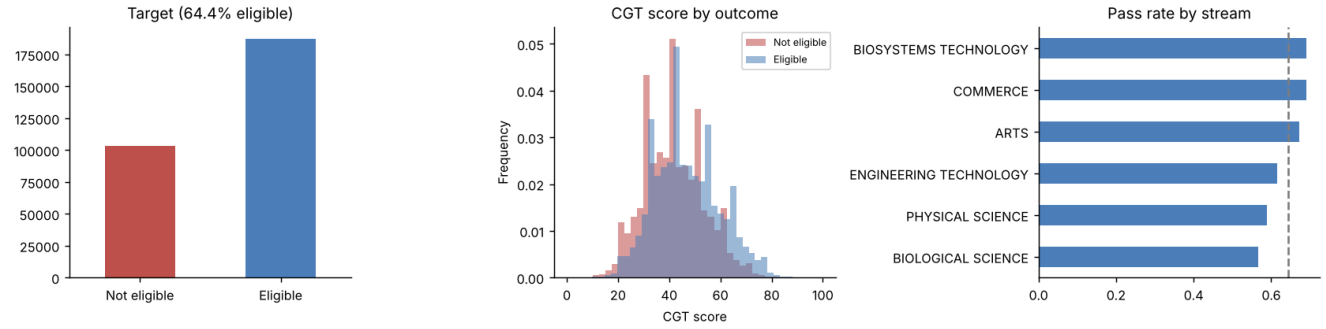


Figure 1. Target balance, CGT score by outcome, and pass rate by (derived) stream.

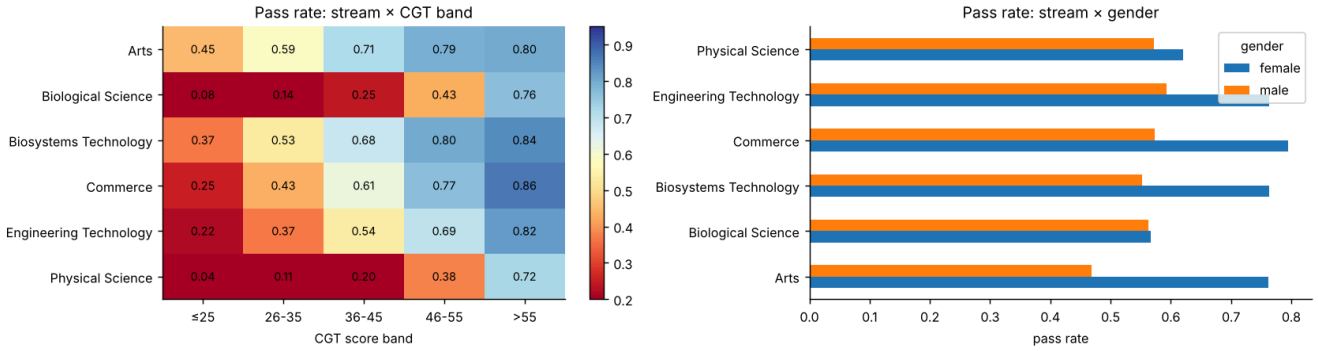


Figure 2. The CGT effect depends strongly on the stream (left), and the gender gap varies by stream (right). Both are non-linear interactions.

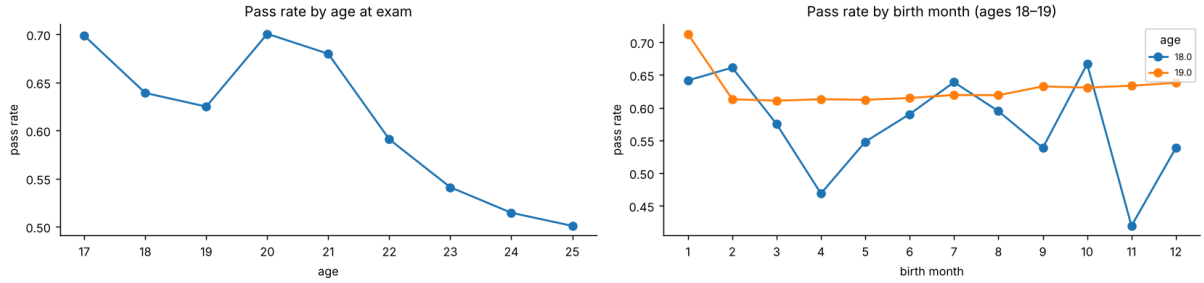


Figure 3. Age is non-monotonic (left). Among 18–19-year-olds, January-born candidates pass more often (right): 71% against 62% at age 19, consistent with the 31 January school-entry cut-off.

1.5 Ethical data use

The data is public, anonymised by its publisher and licensed for non-commercial use, and no attempt was made to identify any person. Even without names, though, combinations of attributes can single people out. A **k-anonymity check** (Table 4) shows that in the public file, 24.6% of candidates are unique on birth date + gender + subject combination. Anyone who knows a classmate's birthday and subjects could look up their grades. This project therefore follows **data minimisation**: day of birth is never used (uniqueness drops to 6.0%), it is stripped from every file shipped with the code and front end. The raw file is **not redistributed** (the repository links to the source instead), and the front end ships only two minimised samples (500 + 5,000 candidates) of the fields the model uses, and only aggregate patterns are reported. Gender is a sensitive attribute, and its use is examined explicitly in §5.

Table 4. Re-identification risk (k-anonymity) for different sets of quasi-identifiers.

Quasi-identifiers	Unique records	Records with k < 5	Median group size k
birth year + month + day + gender + subject combination (public file)	24.6%	42.4%	7
birth year + month + gender + subject combination (used by the model)	6.0%	13.0%	185
age + gender + stream (coarse view)	0.02%	0.08%	16,539

2. Selection of a New Machine Learning Algorithm (15 marks)

2.1 The Explainable Boosting Machine

The lectures covered linear and logistic regression, SVM, decision trees, random forest, AdaBoost, gradient boosting, XGBoost, stacking and voting, naive Bayes and k-means. The algorithm chosen here, the **Explainable Boosting Machine** (Lou et al., 2013; Nori et al., 2019), was not covered. It is a *generalised additive model with pairwise interactions* (GA²M):

$$\text{logit } P(\text{eligible} \mid x) = \beta_0 + \Sigma_j f_j(x_j) + \Sigma_{(i,j)\in\mathscr{I}} f_{ij}(x_i, x_j)$$

Each **shape function** f_j is an arbitrary learned curve for one feature, and $\mathscr{I}$ is a small set of automatically selected feature pairs. Training uses **cyclic gradient boosting**. In each round, the algorithm visits the features one at a time and fits a tiny tree (here 2 leaves) to the current residuals *using only that feature*, with a very low learning rate (0.031). Thousands of such rounds build each f_j gradually, and because only one feature is ever used at a time, the result stays additive. The **FAST** algorithm then ranks all candidate pairs by how much residual they explain, and the top 80 pairs get their own two-dimensional functions f_{ij}. Finally, 8 models trained on bootstrap samples (**bagging**) are averaged, which smooths the curves and gives uncertainty bands. The model is fully transparent: a prediction is *exactly* the intercept plus the sum of the term values, so every prediction decomposes into contributions that can be read off a plot.

2.2 Why the EBM was selected

- **The data called for it.** The EDA showed strong non-linear effects (CGT, age) and pairwise interactions (stream × CGT, gender × stream) but weak correlation for any single feature. Linear models cannot capture this, and boosted trees can but are opaque. The EBM captures non-linear main effects and pairwise interactions while staying interpretable.
- **The application needs explanations.** Teachers acting on a flag need to know *why* a student is at risk, and decisions about young people call for models whose logic can be inspected and audited (Rudin, 2019). With an EBM, the explanation *is* the model, not a post-hoc approximation.
- **The accuracy cost is small.** EBMs typically match gradient-boosted trees on tabular data (Caruana et al., 2015; Nori et al., 2019). This is tested directly in §3.
- **It can be edited.** Because each term is separate, a term such as gender can be inspected, removed or constrained, which supports the fairness analysis in §5.

2.3 How the EBM differs from standard models

Table 5. The EBM compared with standard models.

	Decision tree	Logistic regression	k-NN	RF / XGBoost	EBM (this work)
Model form	single tree of if-then splits	linear in features (log-odds)	majority vote of nearest neighbours	sum of hundreds of deep trees (black box)	**sum of one learned function per feature + selected pairwise functions**
Non-linearity	yes (piecewise)	no (unless engineered)	yes (local)	yes	yes, per feature (shape functions)
Interactions	implicit, unstable	only if added by hand	implicit	implicit, unlimited order	explicit, only pairwise, automatically detected (FAST)
How it is trained	greedy recursive splitting	maximum likelihood	no training (memorises)	bagging / gradient boosting of trees	cyclic gradient boosting: tiny trees on one feature at a time, low learning rate, many rounds, bagged
Interpretability	readable only if shallow	global coefficients	none	needs post-hoc SHAP/LIME	**exact, built in**: each term can be plotted; predictions decompose exactly
Typical accuracy on tabular data	low–moderate	moderate	moderate	high	close to GBDT

Compared with a **decision tree**, the EBM never splits on several features in one path, so it cannot hide long, unstable interaction chains, and it boosts thousands of small steps instead of making one greedy pass. Compared with **logistic regression**, it keeps additivity on the log-odds scale but replaces each linear coefficient with a learned curve and adds pairwise terms, so non-linearity no longer has to be engineered by hand. Compared with **k-NN**, it learns a compact global model (0.2 MB) instead of memorising the data, and it predicts about 200× faster. Compared with **random forest / XGBoost**, it gives up unrestricted high-order interactions in exchange for exact, global and local interpretability.

3. Model Training and Evaluation (20 marks)

3.1 Train / validation / test split

One real candidate was set aside first for the inference demonstration. The rest were split **70 / 10 / 20** with stratification on the target (Table 6). The **training** set was used for fitting and for hyperparameter search. The search used 3-fold stratified cross-validation on a 60,000-row stratified subsample, to keep it tractable on a laptop. The **validation** set was used only to choose the early-warning threshold, and the **test** set was touched once, for the final evaluation. The EBM additionally holds out 15% of each bag internally for early stopping.

Table 6. Data split.

	Rows	Eligible rate
Training (70%)	203,198	64.43%
Validation (10%)	29,028	64.43%
Test (20%)	58,058	64.43%

3.2 Hyperparameter choices

No model uses library defaults. Each was tuned with a randomised or grid search over a justified space, maximising ROC-AUC (Table 7). For the EBM, the space and the reasoning are shown in Table 8. The search preferred the largest number of interactions (80), consistent with the interaction effects seen in the EDA, and a large minimum leaf size (50), a strong regulariser that suits the noisy outcome of a single examination. The final EBM uses 8 bags. The same configuration was also trained **without gender** for the fairness analysis.

Table 7. Selected hyperparameters and cross-validated ROC-AUC (training subsample).

Model	Selected hyperparameters (none left at default)	CV ROC-AUC	Fit (s)
Logistic Regression	C=0.5	0.7432	4
Decision Tree	criterion=gini, max_depth=14, min_samples_leaf=91	0.7474	2
k-NN	n_neighbors=101, weights=uniform	0.7485	0
Random Forest	max_depth=16, max_features=0.5, max_samples=0.9711, min_samples_leaf=18, n_estimators=391	0.7605	84
XGBoost	colsample_bytree=0.7837, gamma=1.6685, learning_rate=0.0278, max_depth=5, min_child_weight=22, n_estimators=508, reg_lambda=8.706, subsample=0.933	0.7692	8
LightGBM	colsample_bytree=0.9826, learning_rate=0.0518, max_depth=-1, min_child_samples=60, n_estimators=356, num_leaves=29, reg_alpha=0.0041, reg_lambda=0.0012, subsample=0.7694, subsample_freq=1	0.7679	4
EBM	interactions=80, learning_rate=0.0313, max_bins=1024, max_leaves=2, min_samples_leaf=50, smoothing_rounds=0, outer_bags=8	0.7667	135

Table 8. EBM search space and rationale.

Hyperparameter	Search space	Selected	Rationale
interactions	20, 40, 80	80	number of pairwise terms; EDA showed stream × CGT and gender × stream effects
learning_rate	0.005–0.05 (log)	0.0313	small steps give smooth, stable shape functions
max_bins	256, 512, 1024	1024	resolution of numeric shape functions (CGT has ~90 distinct values)
max_leaves	2, 3	2	size of each tiny tree in the cyclic booster
min_samples_leaf	2, 10, 25, 50	50	minimum students per leaf: regularises noisy bins
smoothing_rounds	0, 50, 200	0	initial random-split rounds for smoother curves
outer_bags	4 (search) → 8 (final)	8	bagging: averages 8 EBMs and gives the uncertainty bands

3.3 Performance metrics

ROC-AUC is the primary metric: it measures how well a model *ranks* students by risk, independently of any threshold, which suits an early-warning list. Because the at-risk class is the minority (35.6%), **PR-AUC for the at-risk class** and **balanced accuracy** are also reported, along with **macro-F1** and accuracy at the 0.5 threshold. The **Brier score** measures calibration, which matters because teachers see probabilities. 95% confidence intervals come from 300 bootstrap resamples of the test set, and differences between models are tested with a **paired bootstrap** on ΔAUC.

3.4 Results

Table 9. Test-set results (58,058 unseen candidates).

model	Group	ROC-AUC	AUC CI low	AUC CI high	PR-AUC (at-risk)	accuracy	balanced acc	F1 (macro)	Brier	overfit gap	fit time (s)	ms / 1k preds	size (MB)
LightGBM	Black-box reference	0.7723	0.7684	0.7762	0.6557	0.7290	0.6751	0.6828	0.1804	0.0114	4	10.3	0.41
XGBoost	Taught	0.7705	0.7664	0.7743	0.6529	0.7286	0.6717	0.6796	0.1811	0.0070	8	5.4	0.26
EBM	New algorithm	0.7674	0.7632	0.7714	0.6488	0.7260	0.6691	0.6767	0.1822	0.0050	135	3.2	0.23
Random Forest	Taught	0.7656	0.7614	0.7694	0.6439	0.7243	0.6684	0.6759	0.1829	0.0263	84	31.2	20.78
Decision Tree	Taught	0.7552	0.7507	0.7589	0.6315	0.7182	0.6605	0.6675	0.1865	0.0167	2	1.8	0.05
k-NN	Taught	0.7542	0.7497	0.7578	0.6264	0.7164	0.6509	0.6575	0.1869	0.0080	0	261.9	0.66
EBM (no gender)	New algorithm	0.7447	0.7400	0.7486	0.6255	0.7127	0.6462	0.6523	0.1892	0.0028	135	2.2	0.23
Logistic Regression	Taught	0.7428	0.7390	0.7468	0.6145	0.7089	0.6464	0.6524	0.1905	0.0011	4	3.9	0.00

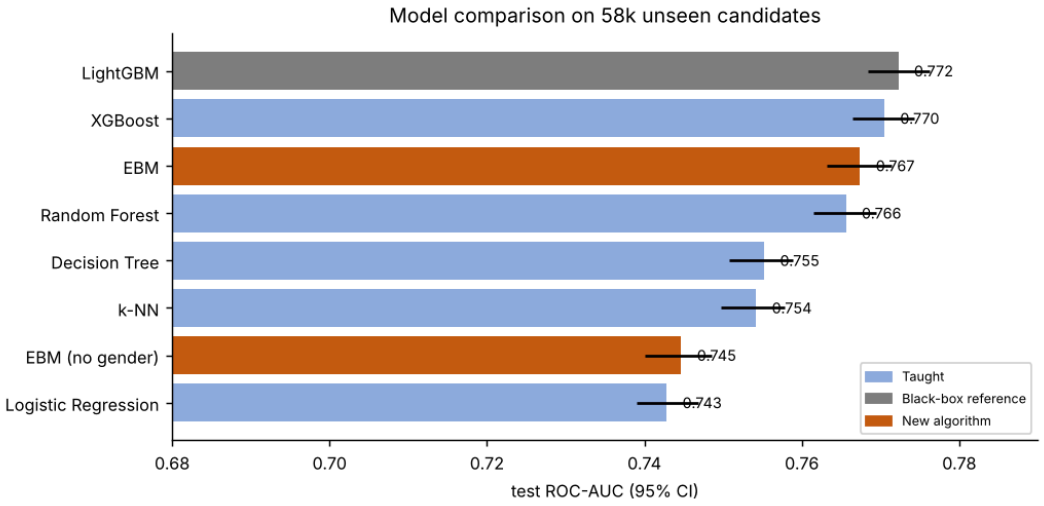


Figure 4. Test ROC-AUC with 95% bootstrap confidence intervals.

Table 10. Paired bootstrap comparison: ΔAUC of the EBM against every other model.

model	ΔAUC (EBM − model)	CI low	CI high	verdict
Logistic Regression	0.0245	0.0224	0.0265	EBM better
Decision Tree	0.0123	0.0106	0.0139	EBM better
k-NN	0.0133	0.0115	0.0149	EBM better
Random Forest	0.0018	0.0005	0.0030	EBM better
XGBoost	-0.0031	-0.0040	-0.0023	EBM worse
LightGBM	-0.0049	-0.0059	-0.0040	EBM worse
EBM (no gender)	0.0228	0.0207	0.0248	EBM better

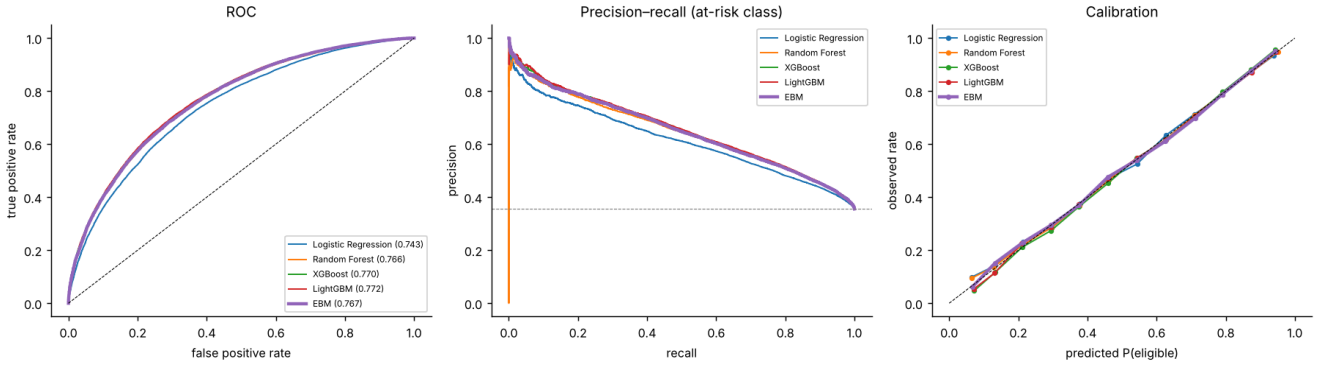


Figure 5. ROC, precision–recall (at-risk class) and calibration curves.

3.5 What the results indicate

- **The EBM significantly beats every interpretable or simple baseline:** logistic regression (+0.025 AUC), the decision tree (+0.012), k-NN (+0.013) and even the random forest (+0.002). All confidence intervals exclude zero.
- **The cost of transparency is small.** XGBoost and LightGBM are ahead by 0.003–0.005 AUC. That gap is statistically significant with 58,000 test students, but it means roughly 1 in 200 pairs of students are ranked differently. In exchange, the EBM gives exact explanations and is the **least over-fitted** model (train–test AUC gap 0.005 against 0.007 for XGBoost and 0.026 for random forest). It is also the fastest to score (3.2 ms per 1,000 students) and small (0.2 MB).
- **It is well calibrated** (Brier 0.1822; the calibration curve follows the diagonal), so its probabilities can be shown to teachers as they are.

- **There is a ceiling of about 0.77 AUC for every model.** This shows that the limit is the information available before results, not the algorithm. School, district, O/L results and socio-economic factors are not in the data (§5).

Early-warning operating point. At the default threshold of 0.5, the EBM catches only 47% of at-risk students. A threshold chosen on the *validation* set (0.66) catches **70%**, with 56% of flagged students actually failing, against a base rate of 35.6% (Table 11).

Table 11. Default vs early-warning threshold (EBM, test set).

setting	threshold	at-risk recall	at-risk precision	flagged %	accuracy	balanced acc
default (0.5)	0.500	0.472	0.661	0.254	0.726	0.669
early warning	0.660	0.696	0.556	0.445	0.695	0.695

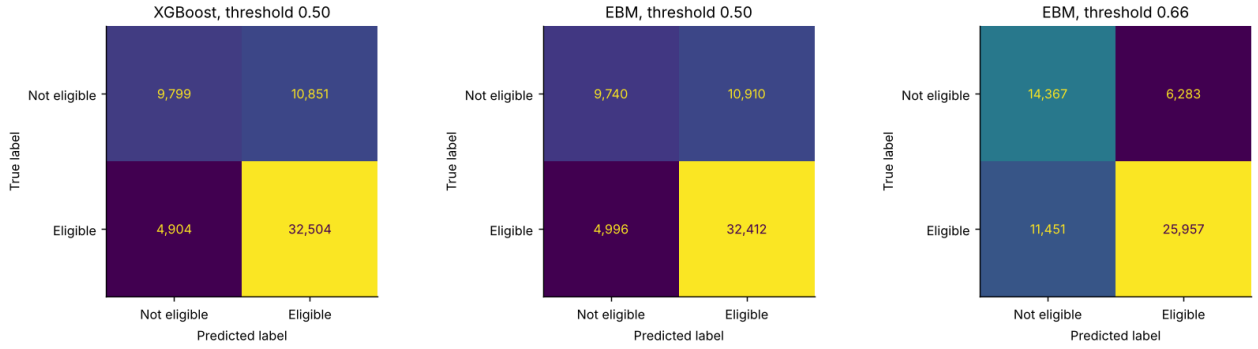


Figure 6. Confusion matrices: XGBoost and EBM at 0.5, and the EBM at the early-warning threshold.

4. Explainability and Interpretation (20 marks)

Five complementary methods were applied: (i) the EBM's own **global shape functions and term importances**, (ii) its exact **local explanations**, (iii) **partial dependence and ICE plots**, (iv) **permutation feature importance**, and (v) **SHAP** on the LightGBM black box and **LIME** on the EBM, used to cross-check the EBM's explanation.

4.1 What the model has learned: global view

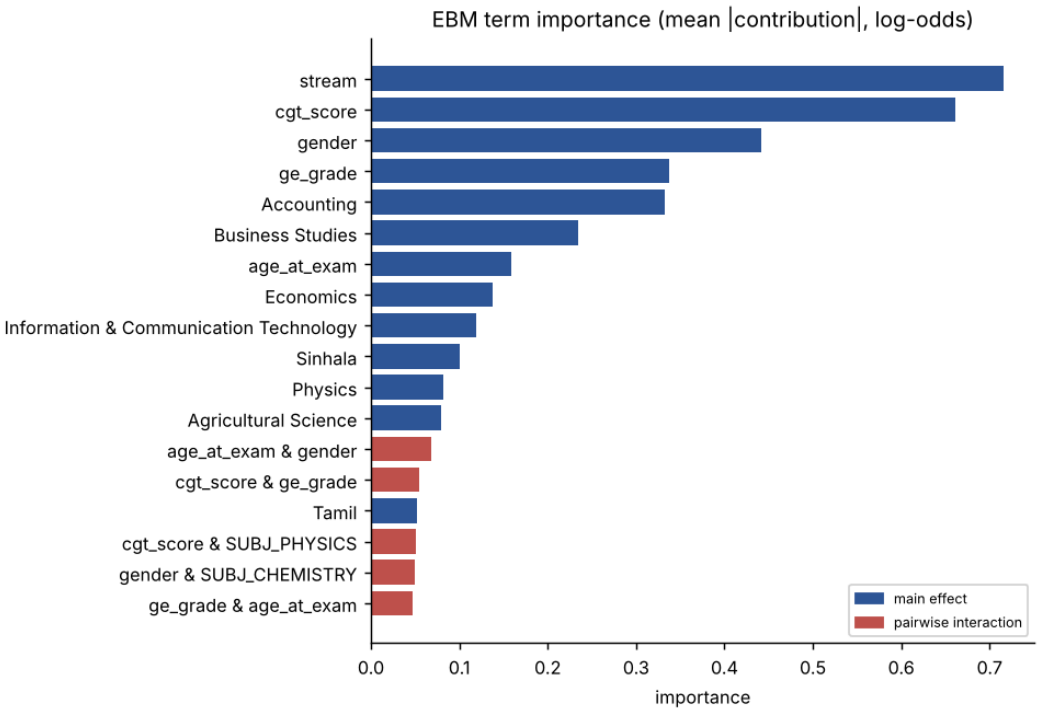


Figure 7. EBM term importance (mean absolute contribution on the test set). Red bars are pairwise interaction terms.

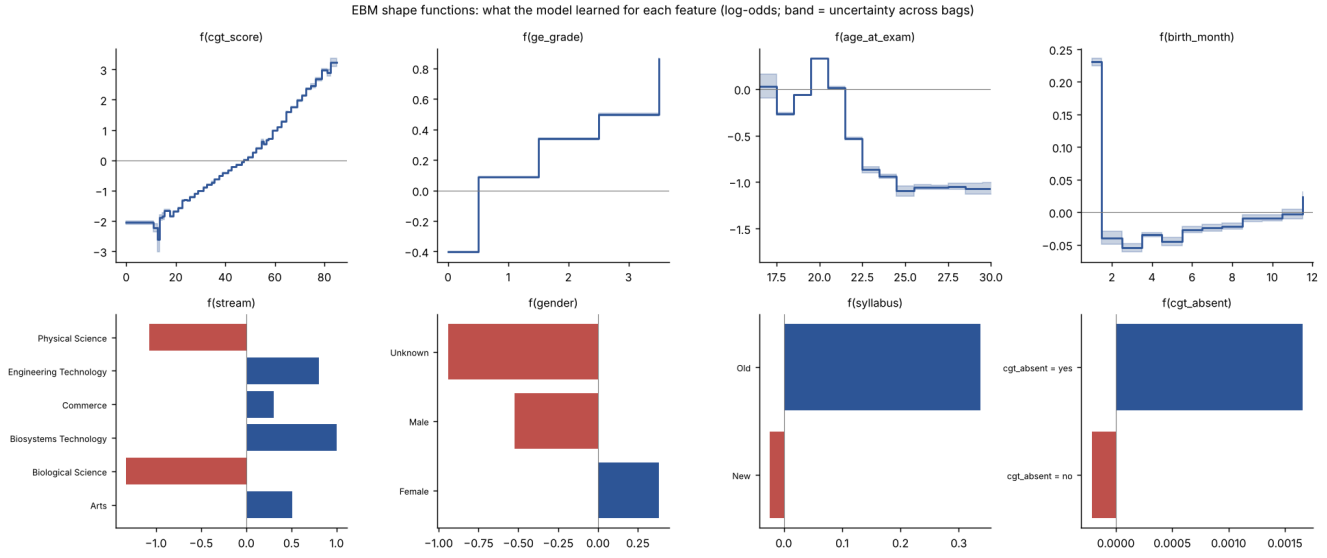


Figure 8. Learned shape functions (log-odds; the shaded band is variation across the 8 bags). These are the model itself, not an approximation.

The **most influential features** are the stream and subject combination, the CGT score, gender and the General English grade, followed by age (Figure 7). The shape functions (Figure 8) show *how* each feature acts:

- **CGT score**: an almost perfectly monotonic, close-to-linear increase from about −2.6 to +3.2 log-odds across the score range. General aptitude is the strongest single signal.
- **General English**: a monotonic step function from F to A (−0.4 to +0.8).
- **Stream**: Biological and Physical Science are strongly negative and the technology streams and Arts positive, matching their pass rates.
- **Age**: a peak at 20 (typically a second attempt), first-timers slightly below zero, and a steep drop after 22.
- **Birth month**: January stands out (+0.23), while the other months are near zero. This is a cohort effect, discussed in §4.6.
- **Missing CGT**: the missing-value bin of the CGT term carries **+0.50** log-odds, which identifies repeat candidates.

Because the stream is a deterministic function of the subjects, the EBM's split of credit between the *stream* term and individual *subject* terms is not unique. The interpretable quantity is their sum for each combination (Figure 9). Science combinations such as Combined Maths–ICT–Physics and Agriculture–Biology–Chemistry carry the largest penalties (-1.89), while Arts combinations in Sinhala, Buddhist Civilisation, Geography and Political Science carry the largest bonuses.

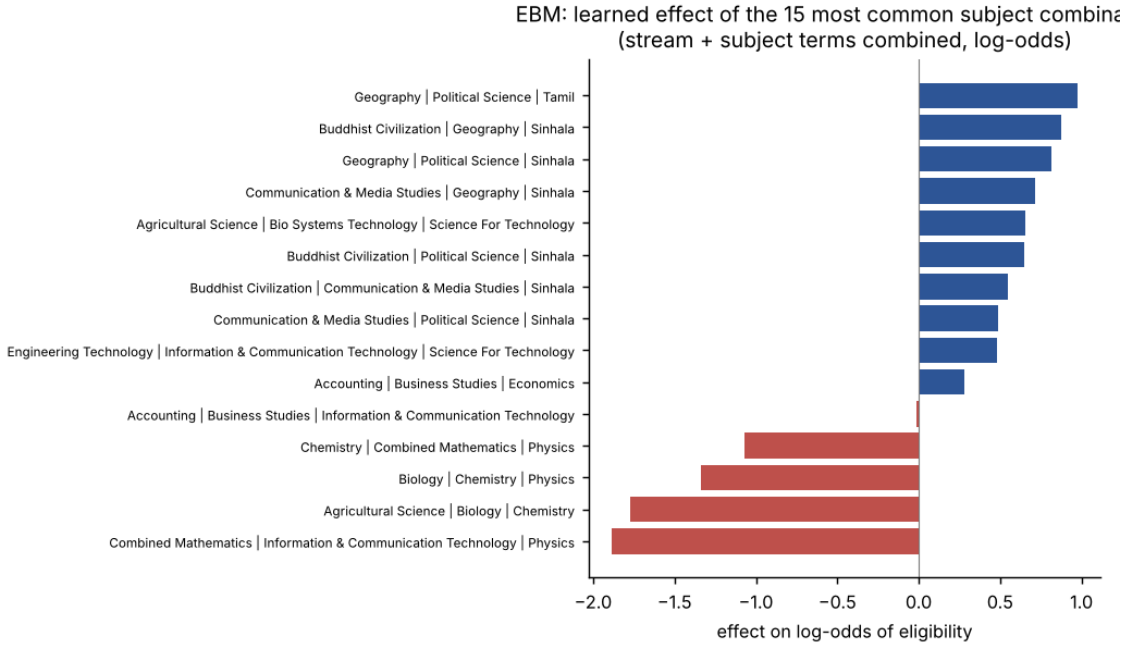


Figure 9. Learned effect of the 15 most common subject combinations (stream + subject terms combined).

4.2 Interactions

The strongest pairwise terms are *Age At Exam & Gender*, *Cgt Score & Ge Grade*, *Cgt Score & Physics*. The **age × gender** term shows that the female advantage shrinks and then reverses with age: among older (repeat) candidates, males do relatively better. The **CGT × General English** term shows the two reinforce each other: a high CGT score is worth more with a good English grade, and an A in English is worth less with a low CGT score. The **CGT × Physics** term shows that a high CGT score is worth *more* for Physics students than for others, consistent with Physics demanding the quantitative reasoning the CGT measures.

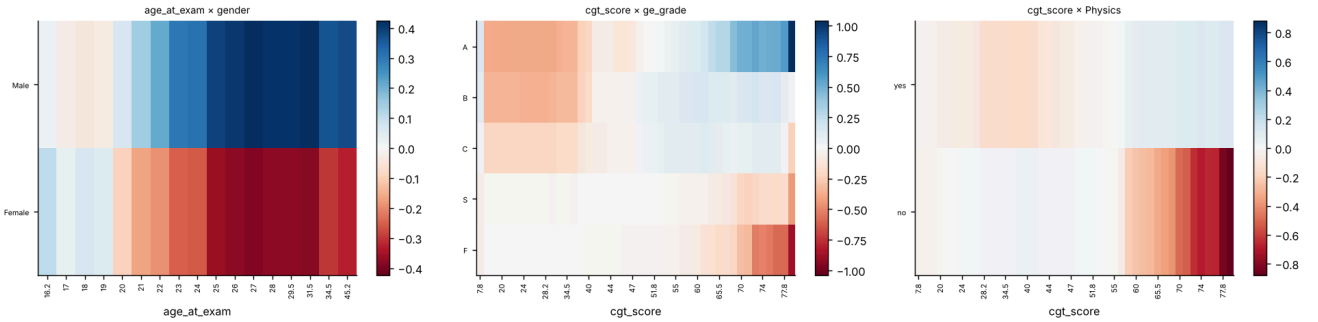


Figure 10. Three of the learned pairwise interaction functions (log-odds).

4.3 Local explanations

An EBM prediction is the intercept (+0.74) plus the sum of its term contributions, so the per-student explanation is exact. Figure 11 explains three students:

- The **held-out candidate** (Arts stream, CGT 43, failed General English, old syllabus) gets P = 0.83, mostly from the Arts stream, being female and the old syllabus, partly offset by the English grade. She did in fact pass.
- The **confidently at-risk** student (Physical Science, CGT 18) gets P = 0.02, driven by a very low CGT score and the science stream.
- The **borderline** student (Commerce: Accounting, Business Studies, Economics; CGT 50; P = 0.66) shows competing forces. Accounting pulls upwards and the other two Commerce subjects pull downwards, so the three should be read together as the combination effect in §4.1. Being male pulls further down, leaving the student just below the early-warning threshold.

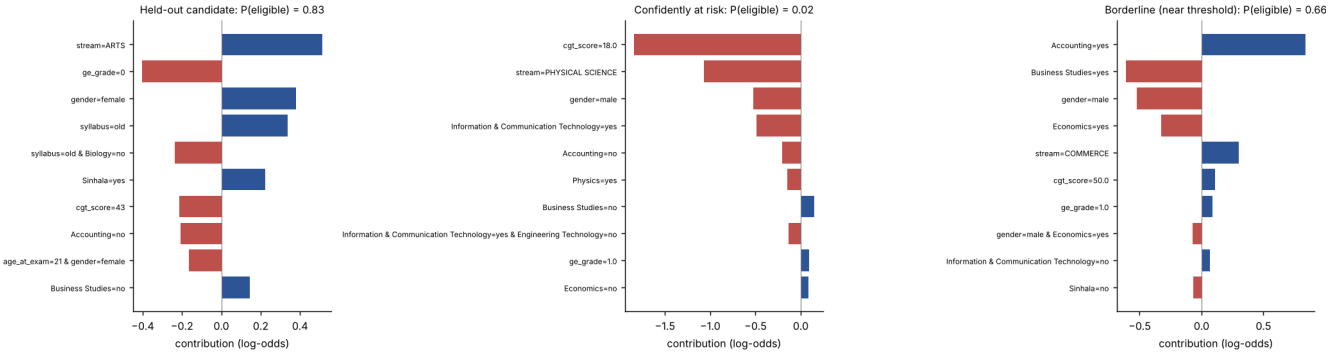


Figure 11. Exact local explanations for three test students (top 10 contributions, log-odds).

4.4 Partial dependence and permutation importance

Partial dependence plots (Figure 12) show the average predicted probability as one feature is varied. For the EBM and the LightGBM black box they are almost identical for CGT, English, age and birth month, so two very different learning algorithms recover the same relationships. The ICE curves (thin lines) are roughly parallel, confirming that the CGT effect is mostly additive. The 2-D PDP (Figure 13) isolates the January × age-18/19 cohort pattern. Permutation importance (Figure 14) ranks CGT, stream, gender and General English highest for the EBM, consistent with its term importances.

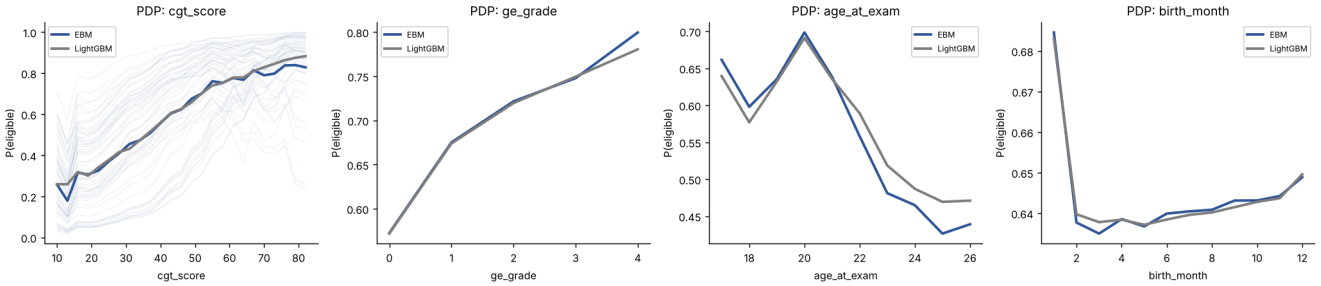


Figure 12. Partial dependence (thick lines) for the EBM and LightGBM; ICE curves for 60 students (thin lines, CGT panel).

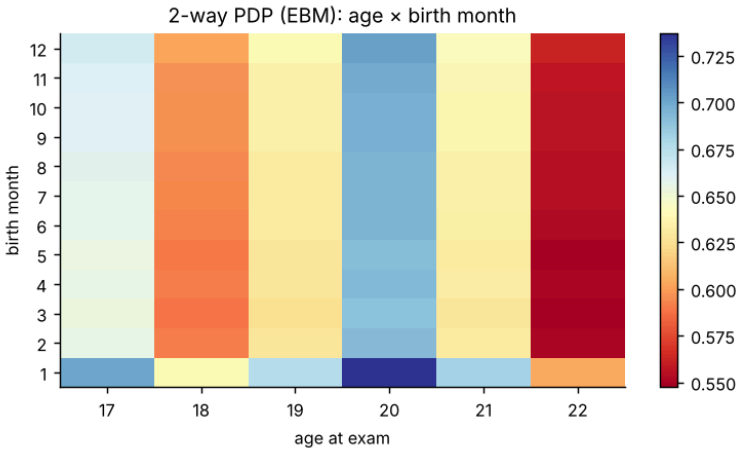


Figure 13. Two-way partial dependence of the EBM: age at exam × birth month.

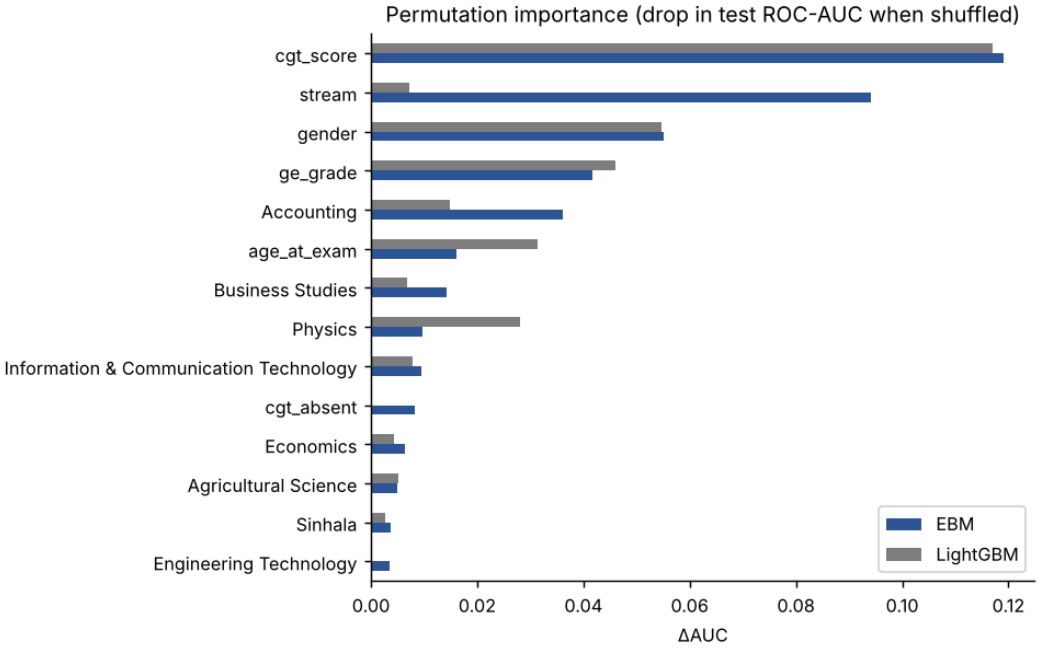


Figure 14. Permutation importance (drop in test ROC-AUC when a feature is shuffled), EBM vs LightGBM.

4.5 Cross-checking explanations: EBM vs SHAP vs LIME

To test whether the explanations are robust, the EBM's exact contributions were compared, in the same 9-feature space, with **TreeSHAP** values of the LightGBM black box (Lundberg & Lee, 2017) and with **LIME** (Ribeiro et al., 2016) applied to the EBM itself, for 200 test students. LIME was configured so that the subject combination is a single feature. Perturbing 43 subject indicators independently would create impossible students, such as one taking nine subjects.

Table 12. Agreement between explanation methods (200 test students).

	Median Spearman ρ	Top-1 match	Mean top-3 overlap (of 3)	Sign agreement
EBM vs LIME(EBM)	0.667	0.665	2.250	0.924
EBM vs SHAP(LightGBM)	0.883	0.765	2.465	0.970

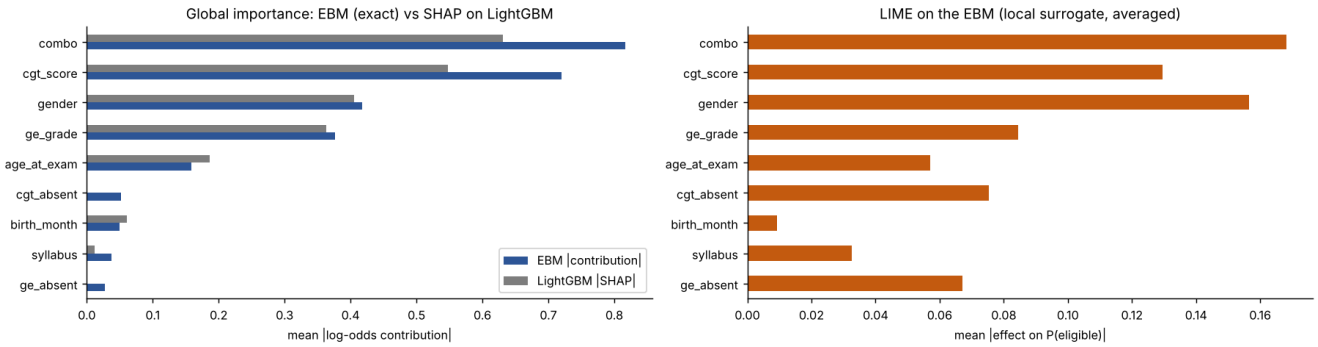


Figure 15. Global importance in a common feature space: EBM (exact), SHAP on LightGBM, LIME on the EBM.

The EBM and SHAP-on-LightGBM agree strongly (median rank correlation 0.88; same top feature for 76% of students; direction agrees 97% of the time). So the glass box and the black box have learned the same drivers. LIME agrees less closely (ρ = 0.67) and is **unstable**: across 8 random seeds, its top-3 features for the same borderline student were identical in only 5 runs. That illustrates why an exact, built-in explanation is preferable to a sampled approximation when one is available.

4.6 Does the model's behaviour align with domain knowledge?

Table 13. Alignment of the learned model with domain knowledge about the Sri Lankan A/L system.

Domain knowledge / expectation	What the EBM learned	Aligned?
Higher general aptitude (CGT) means a higher chance of passing	f(CGT) rises with score (Spearman ρ = 1.00)	✔ yes
English proficiency supports performance	f(General English) increases from F to A (ρ = 1.00)	✔ yes
Science streams are the hardest to pass all three subjects (DoE statistics)	stream effect: science -1.20 vs Arts/Commerce +0.41; Physics/Chemistry/Agri Sci terms negative	✔ yes
School-entry cut-off is 31 January, so January-born students belong to the year-earlier cohort (often repeaters at the same age)	f(birth month = Jan) = +0.231 vs mean of other months -0.022; age × month interaction	✔ yes
Repeat candidates are exempt from the CGT and General English, and repeaters pass more often	f(CGT missing) = +0.50 log-odds	✔ yes
Repeat candidates around 20–21 do better; much older candidates do worse	f(age) peak near 20; f(age ≥ 23) = -1.18	✔ yes
Female candidates outperform males at A/L (also visible in this data: 71.5% vs 54.7%)	f(gender): female +0.38, male -0.52	✔ yes

All seven checks agree. Two are worth highlighting because the model discovered them *without being told*. Grade 1 admission in Sri Lanka requires a child to be five years old by **31 January**, so January-born students belong to the year-earlier school cohort. At the same nominal age they are more likely to be second-attempt candidates, who pass more often: at age 19, 71% of January-born candidates passed against 62% of the rest. Likewise, the positive missing-CGT bin reflects the rule that repeat candidates are exempt from the CGT and General English. A model that learned spurious patterns would not line up with institutional rules like these.

5. Critical Discussion (10 marks)

5.1 Limitations of the model

- **Modest discrimination.** An AUC of 0.767 means about one pair in four is ranked the wrong way round. Even at the early-warning threshold, 30% of at-risk students are missed and 44% of flagged students pass anyway. The tool can prioritise support but must not decide anything on its own.
- **One year only.** 2020 was disrupted by COVID-19, and with no later year available the model has no temporal validation. Patterns may shift from year to year.
- **Timing.** The CGT and General English papers are sat during the A/L examination period, so the warning comes weeks, not months, before results. An earlier model would need O/L results or school-based assessments.
- **Additivity.** The EBM allows only pairwise interactions. The black boxes' 0.005 AUC advantage comes from higher-order effects the EBM cannot represent.
- **Sparse regions.** For very old candidates (over 30) or rare subject combinations, shape functions rest on few students and their uncertainty bands widen accordingly (Figure 8). Predictions there are less reliable.

5.2 Data quality issues

- Community-compiled, not an official release. It covers about 96% of candidates, but the coverage checks (§1.2) show no detectable bias.
- NIC decoding errors affected 0.24% of birth dates and genders, and 32 implausible ages (e.g. 9 or 108) were set to missing.
- The derived stream approximates the official one for the 'NON' (non-standard combination) group.
- Important predictors are absent: school, district, medium of instruction, O/L results, attendance and socio-economic status. This is the main reason all models plateau around 0.77 AUC.

5.3 Risks of bias and unfairness

Gender is the third most important feature: female candidates passed at 71.7% against 54.4% for males, which is also visible in UGC statistics. Using it improves accuracy (+0.023 AUC), but it changes **who gets help** (Table 14, Figure 16). With gender, the model flags 67% of boys against an actual fail rate of 45%, and catches 83% of at-risk boys but only 55% of at-risk girls. At-risk girls are under-detected because the model expects girls to pass. **Without gender**, at-risk recall becomes almost equal (69% for girls, 69% for boys), satisfying *equal opportunity*, at the cost of more false alarms for girls and a 0.023 lower AUC. For a supportive tool whose purpose is to reach every struggling student, the gender-free model is the more defensible default, and the front end offers both. Subgroup AUC is stable across streams (0.72–0.79) and the model is well calibrated within every stream and both syllabi.

Table 14. At-risk detection by gender at the early-warning threshold.

group	model	n	actual fail %	flagged %	at-risk recall	AUC	mean predicted	actual pass %
gender = female	EBM	33835	0.284	0.286	0.545	0.763	0.720	0.716
gender = female	EBM (no gender)	33835	0.284	0.426	0.692	0.757	0.662	0.716
gender = male	EBM	24113	0.455	0.666	0.827	0.734	0.543	0.545
gender = male	EBM (no gender)	24113	0.455	0.512	0.690	0.723	0.623	0.545

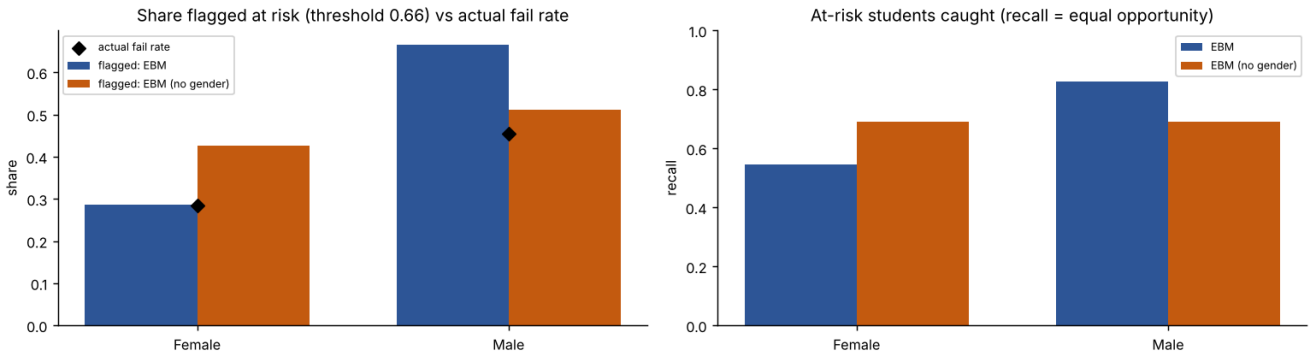


Figure 16. Share flagged vs actual fail rate (left) and at-risk recall (right), by gender, with and without the gender feature.

5.4 Real-world impact and ethical considerations

- **Intended use:** helping schools target *supportive*, *low-cost* interventions such as extra classes, mentoring, or advice on re-sitting and vocational pathways. **Never** admission, streaming or any decision that limits a student's options. A risk label is itself a sensitive inference and can become self-fulfilling if teachers lower expectations.
- **Human in the loop:** the explanation shown with every prediction lets teachers judge whether the reasons make sense for the individual student.
- **Privacy:** day of birth and identifiers are never used, the raw data is not redistributed, and a school using the tool would need consent and data-protection compliance under Sri Lanka's Personal Data Protection Act, No. 9 of 2022.
- **Transparency and accountability:** the glass-box model can be audited term by term, and harmful terms can be removed, which is not possible with a black box.

6. Front-End Integration (bonus, 10 marks)

A **Streamlit** web application (`app.py`) loads the trained pipelines and offers five tabs:

1. **Check a student:** enter the subjects, CGT score (or "did not sit"), English grade, birth year/month, gender and syllabus, and get P(eligible), an at-risk flag, every model's prediction, and an explanation chart. The chart shows the EBM's exact contributions, SHAP for the boosters, or LIME for any model.
2. **Check a class:** upload a CSV of students, get a ranked risk list (scored automatically if results are present) and download it.
3. **Model comparison:** the metrics, confidence intervals, significance tests, curves and tuned hyperparameters.
4. **What the model learned:** the shape functions, interactions, PDPs, method agreement and the domain-alignment table.
5. **Ethics and model card:** intended use, data minimisation, limitations and fairness results. The sidebar switches between the EBM and the gender-free EBM and adjusts the flagging threshold.

Deployment. The app is packaged as a Docker image (`Dockerfile`). A GitHub Actions workflow builds and publishes it to the GitHub Container Registry on every push, and it is hosted on **Azure Container Apps**, following the pattern of the course MLOps workshop, at https://al-early-warning.redwave-d9ae0d05.eastasia.azurecontainerapps.io (East Asia region; it scales to zero when idle, so the first load takes up to a minute). The deployed app works on the **real 2020 data**. Its sample class is 500 held-out test candidates with their actual results, so the app scores itself live, and LIME uses 5,000 real training candidates as background. Both files are minimised as described in §1.5.

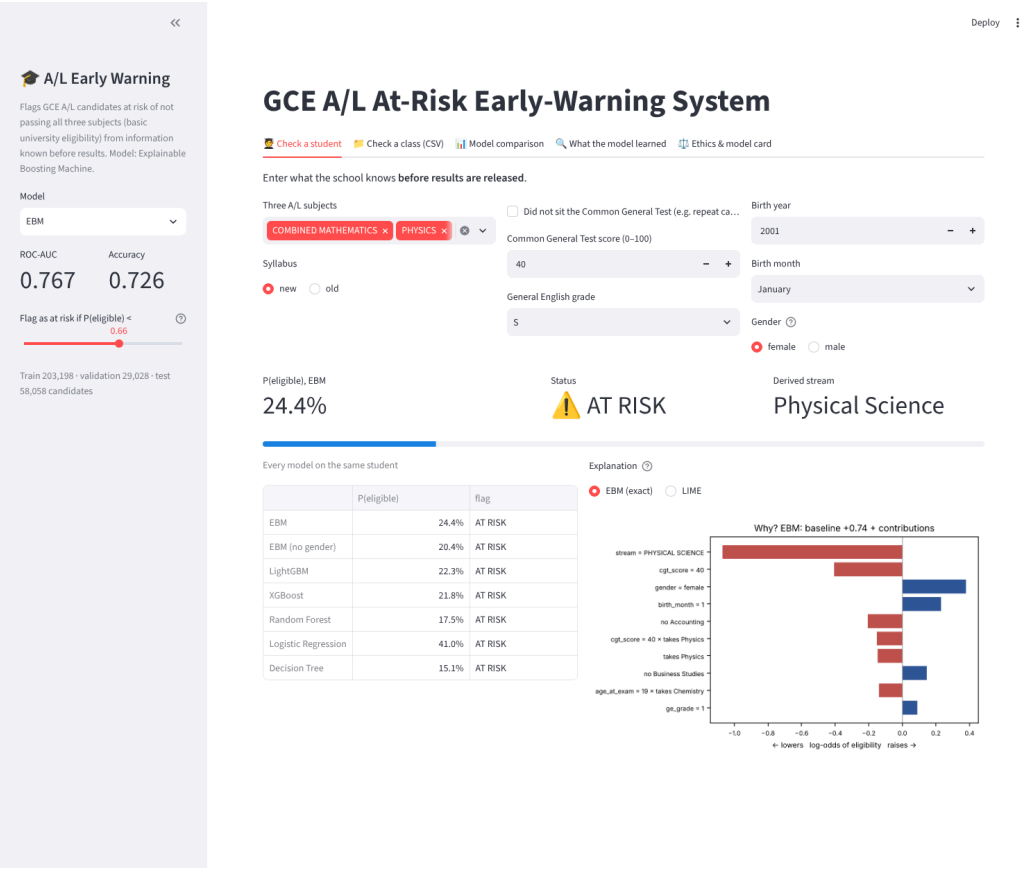


Figure 17. Check-a-student tab with the EBM's exact explanation.

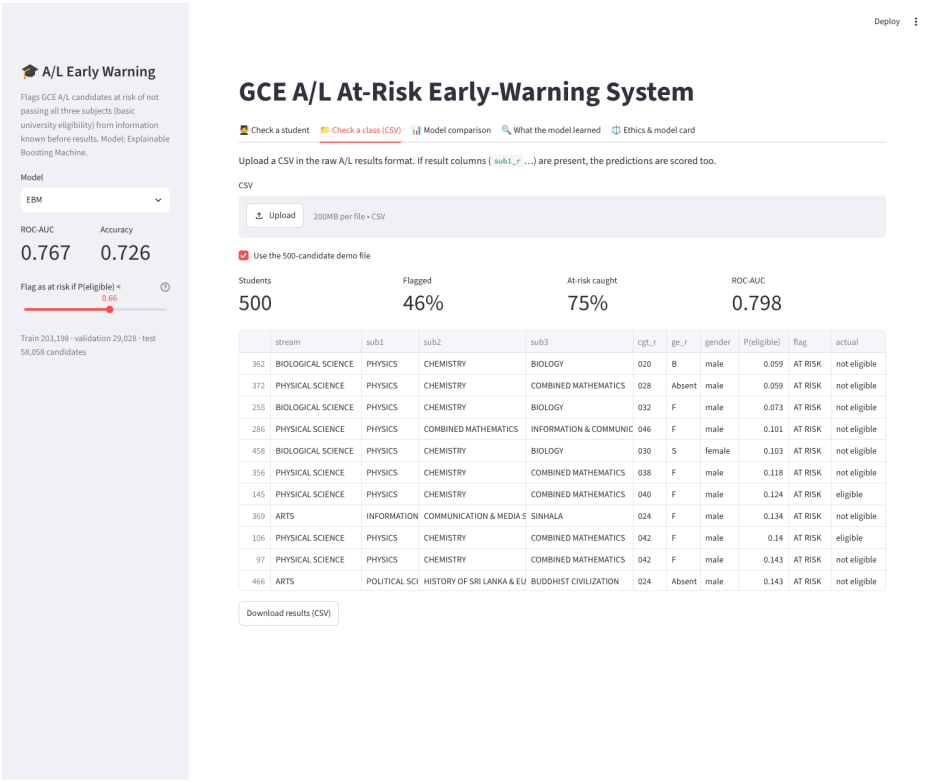


Figure 18. Check-a-class tab: ranked risk list for an uploaded CSV, scored against actual results.

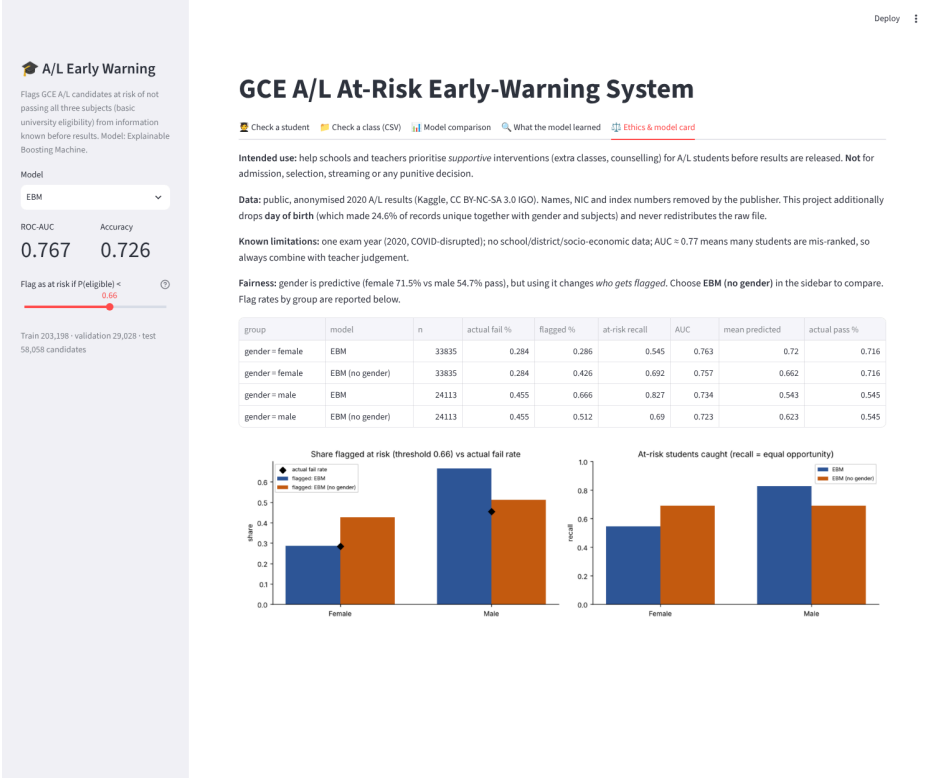


Figure 19. Ethics and model-card tab with the fairness comparison.

7. Conclusion

Using the full national 2020 A/L results, an Explainable Boosting Machine predicts which candidates will fail to meet the basic university requirement with ROC-AUC 0.767. It significantly outperforms the taught interpretable and ensemble baselines and comes within 0.005 of the best black-box gradient-boosted trees, while being fully transparent, well calibrated and the least over-fitted model. Its learned behaviour matches what is known about the A/L system,

including subtle institutional rules it was never told about, and the explanations are consistent across EBM, PDP, permutation importance and SHAP. The main open issues are the information ceiling of pre-results data and the fairness trade-off around gender. Natural next steps are multi-year validation and linking school-level data.

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Appendix A. Comparison with existing Sri Lankan and educational data-mining studies

Published results come from different exams, targets and features, so they cannot be compared one-to-one. Their *methods* were therefore re-implemented on this data (Table 15). Studies reporting over 90% accuracy typically use in-course marks, which are close proxies for the outcome. This work deliberately uses only pre-results information.

Table 15. Methods used in earlier studies, evaluated on this dataset.

Study	Context	Method / finding	Same method on this data
Rajapakse & Ranawaka (2020)	Sri Lanka, Grade 5 scholarship exam, 500 students	Logistic regression	LR: AUC 0.743
Abeywardhane & Gamage (2023)	Sri Lanka, A/L stream recommendation	Random forest best (72% acc.)	RF: AUC 0.766
Bujang et al. (2021)	Malaysia, course grades, 1,282 records	RF best (with SMOTE)	RF: AUC 0.766
Batool et al. (2023), survey	Educational data mining review	DT, NB, ANN, RF most used	DT: AUC 0.755

Appendix B. Reproducibility and tools

Code: `01_preprocess.py` → `02_train.py` → `03_evaluate.py` → `04_explain.py` (about 45 minutes on a 2-core laptop), with `walkthrough.ipynb` as an executed record, `app.py` as the front end, and `report/build_report.py` to generate this report from the result files. Libraries: interpret 0.7 (EBM), scikit-learn 1.9, XGBoost 3.4, LightGBM 4.7, SHAP 0.54, LIME 0.2, Streamlit. Random seed 42 throughout. An AI coding assistant was used to help write and debug code and to generate the front end. All modelling decisions, analysis and interpretation were reviewed by the author.